

Velocity Dispersion During the Milky Way Andromeda Merger

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ABSTRACT

We study the evolution of velocity dispersion during a galaxy merger event. Understanding how velocity dispersion changes during a galaxy merger can help predict how the resulting shape of a galaxy is affected by a merger. The study utilizes an N-body simulation of the merging of the Milky Way (MW)–M31–M33 system. This project predicts Milky Way evolution through velocity dispersion with high-resolution models of the merging event. For each snapshot of the simulation analyzed we find that the velocity dispersion near the edge of the Milky Way the velocity dispersion increases (19.5, 15.2, 18.5, 16.3 km/s at 30 kpc from the MW center for four different snapshots). This conclusion helps to predict the evolution of MW as M31 approaches and eventually collides.

Key words: Velocity Dispersion – Jacobi Radius – Stellar Disk – Stellar Bulge – Local Group

1 INTRODUCTION

This study looks into how velocity dispersion changes throughout the interaction/collision between the Milky Way (MW) and the Andromeda Galaxy (M31). **Velocity Dispersion** is the measure of how randomly stars are moving, a smaller dispersion means stars are moving similarly to one another. These two galaxies are in our **local group**, a collection of many large nearby galaxies as well as the Milky Way’s satellite galaxies. During the MW-M31 merger there are many masses interacting with each other, causing changes in trajectories, which in turn cause velocity changes. Understanding how the velocity dispersion changes during the interaction helps both in predicting the resulting galaxy from the merger as well as understanding how mass is distributed.

Understanding how the velocity distribution changes throughout the collision helps us to understand how galaxies evolve when they interact with their neighbors. A **galaxy** is a gravitationally bound collection of stars whose properties cannot be explained by a combination of baryons and Newton’s laws of gravity. [Willman & Strader \(2012\)](#) Various forces, such as gravity, influence the velocity dispersion, which describes a galaxies evolution whether that be structure, density, or size. **Galaxy evolution** is the process by which a galaxy changes with respect to time, it can be influenced by its stellar population as well as galaxy mergers as shown. Developing a better understanding of how velocity dispersion changes within a galaxy during a merger can help us understand how a galaxy might evolve moving forward, and in turn how a galaxy could have evolved to where it is today.

When two galaxies collide, the large disturbance in each of their systems causes a change in stellar movements. The velocity dispersion is a way to measure the movement of stars through a galaxy. When there is a smaller dispersion of velocities in a system it tells us that all the stars are moving in a more ordered pattern, this could

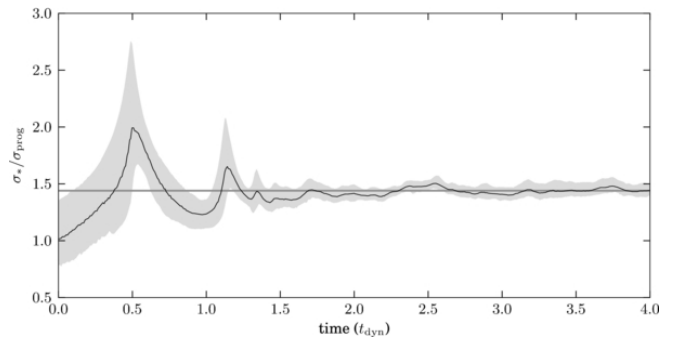


Figure 1. Shows how velocity dispersion changes throughout a merger. Black line is the mean, with uncertainty within the shaded gray region. The gray line is the final mean value of the dispersion.

[Stickley & Canalizo \(2012\)](#)

explain spiral arms. However, with a greater dispersion the star movement looks more random or chaotic, which is more representative of an elliptical galaxy or the bulge of a disc galaxy which is relatively spherical. When two galaxies merge there are forces that drastically change speeds and directions of where stars are moving, but with each close approach of the two galaxies the stars begin to settle and find a new equilibrium as seen in Figure 1 [Stickley & Canalizo \(2012\)](#). These changes can have an effect on the overall structure of the galaxy as well, for example, a high velocity dispersion supports a thickened disk model [Dalcanton et al. \(2023\)](#).

Many questions on this topic relate to creating more accurate and higher-resolution models to describe and predict velocity dispersion in galaxies [Han et al. \(2024\)](#). To create more accurate models more data is required which is expected of the upcoming completion of the Vera C. Rubin Observatory [Pearson, W. J. et al. \(2024\)](#). Following more accurate models higher resolution models can be constructed

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to consider details such as dust absorption as well as 3D velocity dispersion.

2 THIS PROJECT

In this paper, we will evaluate how the velocity dispersion of the Milky Way evolves throughout the MW/M31 interaction. At various important steps during the interaction we will look into how the velocity dispersion changes throughout the galaxy up to its Jacobi Radius. The **Jacobi Radius** is the radius at which a particle is no longer gravitationally bound to a system. Beyond that radius, particles are no longer bound by a galaxy's gravity. Furthermore, the interesting points during the interaction will be when the two centers of mass are at a local maxima or minima. These points of interest can be compared to the initial velocity dispersion to draw conclusions on galactic evolution throughout the merger.

This project addresses trying to predict galaxy evolution through velocity dispersion with high-resolution models of the merging event.

This question is important to look into because it provides the opportunity to deepen our understanding of galaxy evolution. As we currently understand them, galactic collisions are a driver for stellar evolution. [Schawinski et al. \(2010\)](#) It can be difficult to measure velocity dispersion of other galaxies due to limited resolution, thus high-resolution simulations can help us draw connections between the snapshots of galaxies we see today. This study will help by analyzing high-resolution simulation to present the evolution of velocity dispersion throughout the MW/M31 interaction.

3 METHODOLOGY

This simulation utilized in this study is an N-body smoothed-particle hydrodynamics code, GADGET3. [Springel \(2005\)](#) An N-body simulation is a simulation used to model the motion of N individual particles and their gravitational influence on one another. Each galaxy in this simulation is modeled by three components disk, bulge, and halo. The **stellar disk** is the flat component of the galaxy where spiral arms may form and young stars are born. The **stellar bulge** is the bright central region of a galaxy which is usually spherical. A softening length of 2 kpc is also applied at the center of each of the galaxies in order to avoid particles making nonphysical interactions. [van der Marel et al. \(2012\)](#)

For this study, first, we utilize the low-resolution models produced to develop initial results. Following that, higher resolution models will be used in order to generate more impactful results.

The most important calculation in this question is the velocity dispersion σ . It can be calculated just like a regular standard deviation, first, calculating the mean velocity and then calculating the squared average deviation from the mean in each coordinate direction:

$$V_{mean} = \frac{1}{N} \sum_{i=1}^N v_i \quad (1)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - V_{mean})^2} \quad (2)$$

Where N is the number of particles in one measurement and v_i is the velocity of the i th particle. The code will calculate the velocity

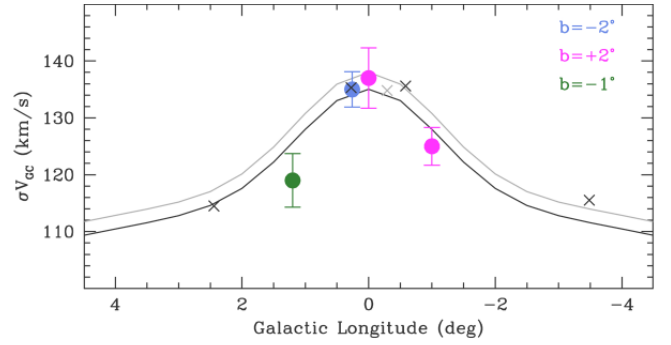


Figure 2. Shows how velocity dispersion changes along the longitude of the Milky Way. The solid lines are the expected radial velocity dispersion trends. While the individual points represent observations at various galactic latitude (b).

[Pearson, W. J. et al. \(2024\)](#)

dispersion of shells of material at constant radii moving outwards towards the Jacobi Radius (R_J). [King \(1962\)](#)

$$R_J = R \left(\frac{m}{2M} \right)^{1/3} \quad (3)$$

Where R is the separation between the two interacting bodies, m is the mass of the host galaxy enclosed by the separation, and M is the total mass of the two galaxies enclosed by the separation. Only Bulge and Disk type particles will be considered because there isn't an observational way to verify the movement of dark matter particles. The code first slices the galaxy into many spherical shells. Within each of these shells eq. 1 will be used to calculate the mean velocity within of bulge and disk particles. Following that eq. 2 will be used to calculate the velocities dispersion within that shell. This process will be repeated for each minima or maxima of the separation between MW and M31.

To answer these questions I will need two distinct plots. The first plot simply shows the separation between MW and M31 highlighting the maxima and minima throughout their interaction. Following that, at each of the minima or maxima I will need to produce a plot that displays the velocity dispersion as a function of radius similar to what is done in Figure 2. [Pearson, W. J. et al. \(2024\)](#) By developing an understanding of how the velocity dispersion changes radially through time we can understand how the shape of a galaxy would change as well.

Based on the work of [Stickley et al. \(2012\)](#), it is suggested by his N-body simulations that the velocity dispersion increases when a galaxy merger occurs. As Figure 1 shows the final equilibrium of the velocity dispersion increases from the value at $t = 0$. I predict that this is happening because of the addition of mass and energy into a system bringing about chaos as the stars attempt to find a new equilibrium. I suspect that following the interaction between the two galaxies the result will be elliptical as the velocity dispersion increases as a whole.

4 RESULTS

As shown in Figure 3 MW and M31 have multiple close approaches before they merge around 6.5Gyr into the simulation. Each of these approaches provide insight into how galaxies evolve when they are disturbed. This figure highlights the important snapshots of the simulation that are the focus of this study. With the two minima occurring at snapshots 55 and 82 and the maxima occurring at snapshots 67

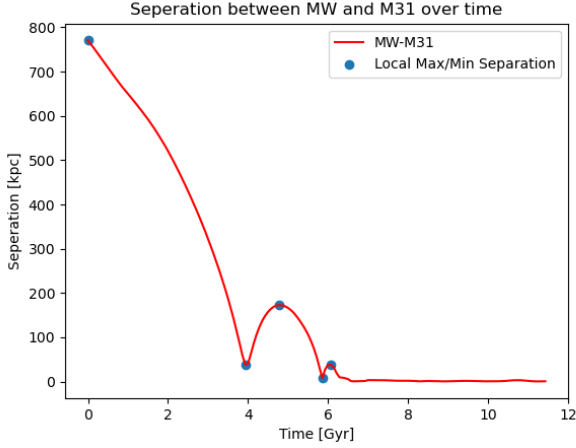


Figure 3. Separation between MW and M31 throughout their interaction. The red line shows the separation between the two galaxies over time. The blue dots depict local maxima and minima throughout their interaction which are points of interest as described in the ‘This Project’ section. The y-axis shows the separation in kpc and the x-axis displays the time during the simulation in Gyr with 0 being present day.

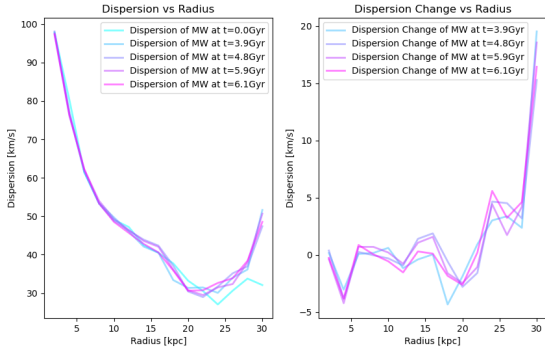


Figure 4. Depicts the velocity dispersion as it changes radially outward from the center of MW on the left. On the right the difference of the velocity dispersion from one snapshot to snapshot 0 is shown. Each continuous line shows how the dispersion changes radially at different times throughout the simulation. The lightest blue represents 0Gyr (today) and as it gets more pink the age progresses to 6Gyr. The y-axis shows the Velocity Dispersion in km/s. The x-axis show the radius in units of Kpc and extends until 30 kpc which is treated as the edge of MW.

and 85. At snapshot 0 when the galaxies are separated by the greatest distance the jacobi radius is similarly large at 485.4 kpc. However, the first minima brings the two bodies very close to each other which shrinks the jacobi radius to 24.6 kpc it continues to follow a damped oscillation at increasing and decreasing at each maxima and minima. The jacobi radius continues to fluctuate going to 109.4 kpc then 5.1 kpc and lastly 24.6 kpc. The separation is important to understand why the MW particles might accelerate or tend to a certain direction which ultimately affects the dispersion. This figure highlights two maxima, two minima and a baseline separation between MW and M31 and serves as a building block for the rest of the study.

In figure 4 we produce interesting results. This is especially visible on the farther end of the radius axis. In the present day line we can

see that the velocity dispersion reaches lower values whereas after the first close approach of the two galaxies the velocity dispersion is higher. However, what is even more interesting is that their interaction before a minima and before a maxima are also different. For a line that represents a minima the line increases from the snapshot that precedes it, and a line representing a maxima slightly decreases from the snapshot that precedes it. Additionally, from one minima to the next minima and from one maxima to the next maxima the velocity dispersion increases. So not only is velocity dispersion changing at each of the interesting snapshots but the velocity dispersion as a whole is trending upwards throughout their interaction even if it decreases at times. Considering only the difference between the first and last snapshot (0-6.071 Gyr) the difference in the dispersion at farther radii (24-30 kpc) is in the range of 5-56 km/s, depending on the radius. At farther radii the velocity dispersion throughout the interaction increases adding to the disorder and chaotic order of particles (or stars) in these regions.

5 DISCUSSION

The velocity dispersion does tend to increase at larger radii. This agrees with the hypothesis that as the two galaxies interact there should be an increase in the velocity dispersion of the galaxy. The velocity dispersion calculated at the latest snapshot compared to today at far radii (24-30 kpc) increase from 5.6 km/s to 16.4 km/s. This is a significant increase in velocity dispersion for the particles that are farther from the center of MW.

The increase of velocity dispersion depicted in figure 4 was similarly shown in figure 1. [Stickley & Canalizo \(2012\)](#) Additionally, the oscillatory motion of the velocity dispersion as it settles into a new equilibrium is also shown in figure 1. [Stickley & Canalizo \(2012\)](#) These results further verify that the motion of stars within a galaxy are greatly perturbed throughout a galactic merger. However, it is also interesting to see that the oscillatory changes in the velocity dispersion suggest the particles finding a new equilibrium as the two galaxies become one.

In the analysis of the data the biggest source of error likely came from neglecting the effects of M33 which is also present in the simulation. M33 has a gravitational influence on the particles in the simulation but was ignored because it does not come close to MW until after the merger. However, M33 is causing some change in the velocities of the particles in MW.

6 CONCLUSION

This study peered into the evolution of velocity dispersion during a galaxy merger event. Understanding how velocity dispersion changes during a galaxy merger can help draw predictions for how the shape of a galaxy is affected by the merger event. The study utilizes an N-body simulation of the merging of the Milky Way (MW)–M31–M33 system. This study has predicted Milky Way evolution through changes in velocity dispersion with high-resolution models of the merging event.

We found that especially at distances farther from the center of the Milky Way the velocity dispersion increases more. At 30 kpc from the center at 4 separate snapshots the velocity dispersion changes by 19.5, 15.2, 18.5, and 16.4 km/s as compared to the velocity dispersion at that distance today. This agrees with the hypothesis proposed earlier that the velocity dispersion will increase however, it is not as severe as

what was predicted. The velocity dispersion most noticeably changes beyond 24 kpc from the center.

Future studies on this topic should be done in order to address the effects of M33. In this study only M31 was considered in calculations however M33 was also a part of the simulation. Additionally, studies into the velocity dispersion of the post merger galaxy should be performed so that comparisons can be drawn between the three stages (current, during the merger, post-merger). Future studies may also want to consider particles that are at first bound to one galaxy but become bound to another galaxy during the interaction.

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We respectfully acknowledge the University of Arizona is on the land and territories of Indigenous peoples. Today, Arizona is home to 22 federally recognized tribes, with Tucson being home to the O’odham and the Yaqui. The University strives to build sustainable relationships with sovereign Native Nations and Indigenous communities through education offerings, partnerships, and community service.

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